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International A Level

Physics (9630)
PH04 Energy and energy resources
Report on the Examination

June 2025

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REPORT ON THE EXAMINATION: INTERNATIONAL A LEVEL PHYSICS (9630) PH04 – JUNE 2025

QUESTION 01

This question was about the cooling curve of a liquid that changes state.

Question 1.1 was the most difficult question on the paper, with less than 20% of students getting a mark. Many students did not spot that the experiment took over 13 minutes to perform. Common answers that did not gain credit referred to the precision or accuracy of digital devices. Some answers gained credit for concurrent readings or close together readings, although it was not particularly relevant in this situation.

Although just half of the students gained a mark for 1.2 for identifying that latent heat is being transferred, very few gained both marks. This required a realisation that, for the temperature to remain constant, the rate of heat transferred from the substance was equal to the rate of energy transfer from latent heat.

The calculation in 1.3 was very accessible, with over 75% of students getting all four marks. The mark scheme allowed credit for the various steps in the calculation, and this meant that it discriminated well. A common error was needlessly attempting to convert temperatures to kelvin, or not taking into account the energy transferred as a solid.

Many students found 1.4 difficult as they struggled to draw a tangent to the curve in Figure 1. This meant that many answers were outside of tolerance. Students should be encouraged to make use of curves like this as part of their practical work. The relatively low discrimination for this question suggests that not all students get these opportunities.

The calculation in 1.5 was quite accessible and discriminated well. A common error was to use the rate of temperature decrease as the power input. The mark scheme allowed these students to gain partial credit provided it was clear what they were doing. Students should be encouraged to set out all of their work carefully so that they do not lose marks through lack of clarity in their answers.

QUESTION 02

This question was about a demonstration of the operation of the lungs and diaphragm.

Question 2.1 was the easiest question in Section A, with over 90% of students gaining both marks. As the question asked about a *change* in energy, students were not penalised for using 3/2 kT with T in °C. It should be noted that future mark schemes may not be this generous.

Question 2.2 was correctly answered by over 75% of students and it discriminated well. Some students had difficulties with the conversion of cm^3 into m^3. Others did not spot that the question asked about one balloon and doubled their answer. Some students tried to work out a change of kinetic energy, despite it being clear in Table 2 that the temperature stayed the same.

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Question 2.3 was another straightforward calculation that was correctly answered by over 70% of students. The mark scheme allowed for partial credit provided the student's answer made it clear what they were doing. Some students had difficulty converting to kelvin or realising which equation they had to use.

Question 2.4 was much more difficult, with less than 10% of students gaining all three marks. Many did not gain the third marking point because their answers did not make it clear that the work done was related to the *change* in the number of particles, rather than the number of particles in the balloon.

QUESTION 03

This question was about the radius of the nucleus.

The calculation in 3.1 was completed correctly by more than 85% of students. The mark scheme rewarded each step in the process independently, so that students who did not get the correct answer still had access to other marks if their working was clear.

Question 3.2 was only slightly more demanding than 3.1. A common missed step was subtracting 26 from their nucleon number. Another common error was to use a value of R_o that they remembered rather than using the data provided in the question. There were several ways of approaching this question that were seen by examiners, and each gained credit.

Question 3.3 was the most discriminating question on the paper, despite it being asked in different forms before. Several students struggled with the charge on the alpha particle, using the number of nucleons rather than the number of protons. Problems working out the mass of the alpha particle were also seen.

Over 60% of students gained the mark in 3.4. To gain the mark, students had to make it clear that it was the *nuclei* of X and Y that had the same charge.

QUESTION 04

This question was about fission and fusion.

Question 4.1 produced a good spread of marks and discriminated reasonably well. Many students did not gain the fourth marking point. Some simply stated that fission released more energy, without giving any explanation or making any reference to Figure 4.

The calculation in 4.2 was much more accessible, with almost 80% of students gaining all three marks. Some student lost marks by leaving their answer in MeV, not including the number of nucleons in their calculations, or including the rest mass of the neutrons. This question discriminated very well.

Question 4.3 was also quite accessible. A common error was to use the same value for the mass of the proton and the mass of the neutron rather than the values given in the data booklet. The structure of the mark scheme meant that partial credit could still be given.

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QUESTION 05

This question was about a practical to determine the moment of inertia of a bicycle wheel.

Both question 5.1 and question 5.2 discriminated very well and produced a good range of marks.

Several students had difficulties converting revolutions to radian in question 5.1. Some students also did not calculate the angular deceleration correctly. Using an energy approach seemed to present students with fewer problems.

The most common difficulty with question 5.2 was the realisation that the angular displacement was not 5°, but 175°. Students who set out their work, and worked through the question logically, had more success.

The description required in 5.3 was much more difficult. Many students simply stated that repeats should be made, despite this having been given in the question. Stating that any anomalous results should be rejected was rarely seen but was given credit. Many students stopped at determining the percentage uncertainty and therefore could not obtain the fifth marking point. The best answers tended to take the alternative route given in the mark scheme.

QUESTION 06

This question was about alternative energy sources.

Question 6.1 was a level of response question about a wind farm. This produced a good spread of marks and discriminated reasonably well. Students should be encouraged to practise questions of this kind and to analyse the mark scheme carefully to help them in future examinations. Students often missed marks by not answering some of the question parts. Answers that lacked detail or were vague also did not gain much credit.

Question 6.2 was answered correctly by more than 80% of students. Some problems were seen with the unit conversions, and with the formula for the surface area of a sphere. This formula is given in the data booklet.

It was clear in question 6.3 that many students had a good understanding of this subject. Again, answers that were vague did not gain credit. Some students also only discussed one area and therefore could not gain more than one mark. Students should be encouraged to use the marks available for a question to show how many points are needed for a full answer.

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SECTION B

About half of the multiple-choice questions were answered correctly by over 80% of the students.

The most difficult question was B19, which just under 60% of students got right. This was the second part of a graph analysis question based on the gas laws. Students who struggled may not have appreciated the effect of temperature on the values of pressure and volume for an ideal gas.

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